EVERY TWO-COLOURING OF THE PLANE CONTAINS A MONOCHROMATIC $T(1,3,4)$

ALINA YAN

Abstract. We prove that every red–blue colouring of the Euclidean plane contains three collinear points of one colour with consecutive distances 1 and 3. The argument works on the triangular lattice $\Lambda = \mathbb{Z}[\omega]$ together with a single rotated copy $q\Lambda$, where $q = \pi/\bar{\pi} = (3+5\omega)/7$ and $\pi = 2+\omega$; the two lattices share a sublattice of index 7. Using the structure theorem for $T(1,3,4)$-avoiding colourings of the triangular lattice from the author's thesis, we show that every lattice line is coloured in runs of exactly four (Rule A) and that the resulting bands never change their offset step (Rule B). Then any two points at distance 8 have the same colour, which forces a monochromatic unit triangle.

1. Introduction

For $a, b, c > 0$ write $T(a, b, c)$ for a configuration of three points with pairwise distances a, b, c, the degenerate (collinear) case allowed, and $T(s) = T(s, s, s)$. A colouring *contains* a configuration if some congruent copy is monochromatic. Erdős, Graham, Montgomery, Rothschild, Spencer and Straus [1] initiated the study of which configurations every two-colouring of the plane must contain. Currier, Moore and Yip [2] proved that every two-colouring of the plane contains a monochromatic three-term arithmetic progression, that is, a monochromatic $T(1, 1, 2)$, which was an inspiration to this research.

Theorem 1.1. *Every two-colouring of $\mathbb{R}^2$ contains a monochromatic $T(1, 3, 4)$.*

Scaling gives a monochromatic $T(a, 3a, 4a)$ for every $a > 0$. The proof has four steps.

(1) *Rule A* (Section 3): along the long direction of any unit triangular lattice, monochromatic runs have length exactly four.
(2) *Bands* (Section 4): every unit triangular lattice is coloured in bands of width four, whose offsets advance by 4 or 5 from row to row.
(3) *Rule B* (Section 5): these steps never change between 4 and 5.
(4) *Distance eight* (Section 6): any two points at distance 8 have the same colour, so every unit equilateral triangle is monochromatic.

The structural input is the description of $T(1, 3, 4)$-avoiding colourings of the triangular lattice from the author's thesis [3], and the idea of overlapping a lattice with a rotated copy proposed in its conclusion. Both finite arguments use only the base lattice and one rotation.

2. Preliminaries

Let $\omega = e^{i\pi/3}$, so $\omega^2 = \omega - 1$ and $\bar{\omega} = 1 - \omega$, and $\Lambda = \mathbb{Z} + \mathbb{Z}\omega$ with $|a + b\omega|^2 = a^2 + ab + b^2$. A *unit triangular lattice* is a set $z_0 + u\Lambda$ with $|u| = 1$; its three *directions* are $u, u\omega, u\omega^2$, and its *rows* in a direction are the lines parallel to it.

Date: October 2026.

The rotation. Let $\pi = 2 + \omega$, a prime of norm 7, and

$$q = \frac{\pi}{\bar{\pi}} = \frac{\pi^2}{7} = \frac{3 + 5\omega}{7}, \qquad |q| = 1.$$

Since $\pi = q\bar{\pi}$, the sublattice $\pi\Lambda$ of index 7 lies in both Λ and $q\Lambda$ (Figure 1). Every point of $\Lambda \cup q\Lambda$ is $[a, b] := (a + b\omega)/7$ with $a, b \in \mathbb{Z}$; it lies in Λ iff $7 \mid a, b$, and in $q\Lambda$ iff $49 \mid 8a + 5b$ and $49 \mid 3b - 5a$. Points of $q\Lambda$ are written $q(m+n\omega)$. The mirror rotation $(5+3\omega)/7$ is the image of q under reflection in the real axis, up to a unit, and could replace q throughout after reflecting every picture.

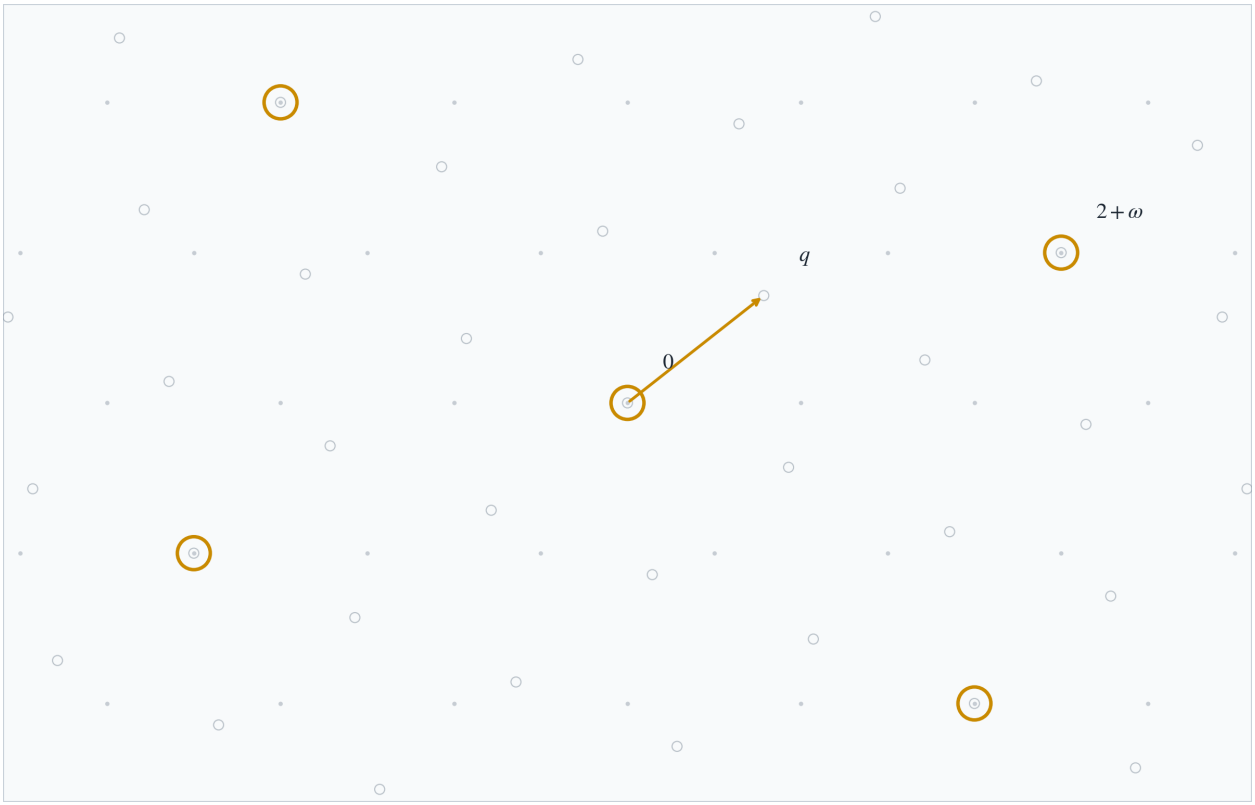


FIGURE 1. Λ (grey dots) and $q\Lambda$ (grey circles); the ringed points form the common sublattice $\pi\Lambda$, which contains 0, $q\bar{\pi} = 2 + \omega$ and its multiples.

Forbidden triangles. A two-colouring of the plane is *admissible* if it contains no monochromatic $T(1, 3, 4)$; we assume one is given and derive a contradiction. The following is the eight-point argument of [1].

Lemma 2.1 (bridge)**.** *An admissible colouring contains no monochromatic $T(1)$, $T(3)$ or $T(4)$.*

Proof. With $c = 1$, $b = 4$ put $A = 0$, $B = c$, $C = b$, $D = c - c\omega$, $E = b + (c - b)\omega$, $F = b - b\omega$, $G = c - b\omega$, $H = b + c - b\omega$. The triples $ABC, ADF, BDG, CEF, FGH, BEH$ are copies of $T(1, 3, 4)$, and $ABD, ACF, BCE, DFG, EFH, BGH$ are equilateral of sides $1, 4, 3, 3, 1, 4$; in particular ABD, BCE, ACF have sides 1, 3, 4. Assume there is no monochromatic $T(1, 3, 4)$, so each of the six collinear triples contains both colours. If A, B, D are red, then ABC, ADF, BDG give C, F, G blue, then CEF, FGH give E, H red, and BEH is red. If A, C, F are red, then ABC, ADF, CEF give B, D, E blue, then BDG, BEH give G, H red, and FGH is red. If B, C, E are red, then ABC, CEF, BEH give A, F, H blue, then ADF gives D red and BDG gives G blue, and FGH is blue. Every equilateral triangle of side 1, 3 or 4 is congruent to ABD, BCE or ACF, and exchanging colours covers blue triangles. □

A *forbidden triangle* is a $T(1)$, $T(3)$, $T(4)$ or $T(1,3,4)$. Every finite argument below uses one rule:

> **(F)** If two vertices of a forbidden triangle have the same colour, the third has the other colour.

The structure theorem. We use the following result from the thesis.

Theorem 2.2 ([3, Theorem 3.13])**.** *In a $T(1,3,4)$-avoiding two-colouring of a unit triangular lattice, one of the three directions, the* long direction*, has all maximal monochromatic runs of length 3 or 4; in each of the other two directions every run has length 1 or 2.*

The restriction of an admissible colouring to any unit triangular lattice is $T(1,3,4)$-avoiding, so Theorem 2.2 applies to Λ, to $q\Lambda$, and to all their images under isometries.

3. Rule A

Theorem 3.1 (Rule A)**.** *In an admissible colouring, no line of a unit triangular lattice carries a maximal monochromatic run of exactly three points.*

Lemma 3.2 (the row)**.** *If such a run exists then, after an isometry and possibly exchanging colours, $0,1,2$ are red and $-3,-2,-1$ and $3,4,5$ are blue.*

Proof. Normalise the run to $0,1,2$ red with $-1,3$ blue, in a lattice Λ. A run of length three lies in the long direction by Theorem 2.2, and runs there have length at least three, so the blue runs through -1 and 3 contain $-3,-2,-1$ and $3,4,5$. □

Lemma 3.3 (the boundary)**.** *In this situation $1-\omega, 2-\omega, \omega, 1+\omega$ are blue, at least one of $-\omega, 3-\omega$ is blue, and at least one of $-1+\omega, 2+\omega$ is blue.*

Proof. The unit triangles $\{0,1,1-\omega\}$, $\{1,2,2-\omega\}$, $\{0,1,\omega\}$, $\{1,2,1+\omega\}$ have two red vertices. The rows $\mathbb{Z} \pm \omega$ are lines of the long direction, so the blue runs through $1-\omega, 2-\omega$ and through $\omega, 1+\omega$ have length at least three. □

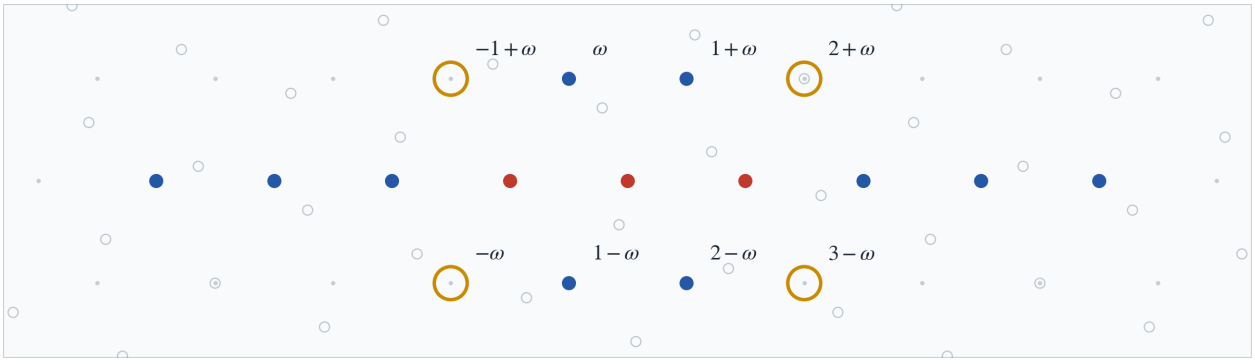


FIGURE 2. Rule A: the row $-3,\dots,5$, the four points forced blue by unit triangles, and the four boundary points (ringed), of which each pair below and above contains a blue point.

Lemma 3.4 (symmetries)**.** *The reflections $\tau(z) = \bar{z}$ and $\sigma(z) = 2-\bar{z}$ preserve the row of Lemma 3.2 with its colours. σ swaps $-\omega \leftrightarrow 3-\omega$ and $-1+\omega \leftrightarrow 2+\omega$; τ swaps $-\omega \leftrightarrow -1+\omega$ and $3-\omega \leftrightarrow 2+\omega$. If χ is admissible, so is $\chi \circ \rho$ for every isometry ρ.*

Proof. Both maps preserve the real axis, τ pointwise and σ as $x \mapsto 2-x$, which preserves the colours of $-3,\dots,5$. For example $\sigma(-\omega) = 2 - \overline{-\omega} = 2-(\omega-1) = 3-\omega$. □

Proof of Theorem 3.1. By Lemma 3.3 the colours of $(-\omega,\ 3-\omega,\ -1+\omega,\ 2+\omega)$ have a blue point in each pair: nine possibilities. The group $\{1, \sigma, \tau, \sigma\tau\}$ permutes them in four orbits, of sizes $1, 4, 2, 2$, with representatives: all four blue; $3-\omega, -1+\omega, 2+\omega$ blue; $3-\omega, 2+\omega$ blue; $-\omega, 2+\omega$ blue. Replacing χ by $\chi \circ \rho$ we may assume the boundary is one of these. Each is refuted inside $\Lambda \cup q\Lambda$ by rule (F), with one case split; the complete deductions are in Appendix A. □

Remark 3.5. A second proof uses only σ: assume $3-\omega$ is blue, then split on the point q. If q is red, rule (F) alone gives a contradiction; if q is blue, one more split finishes. It needs two splits in total but 81 points instead of 49; see Appendix B. In particular q is blue in every such configuration.

4. Bands

Let $h : \mathbb{Z}/8 \to \{\mathrm{R}, \mathrm{B}\}$ be red on $0, 1, 2, 3$ and blue on $4, 5, 6, 7$.

Lemma 4.1 (bands of width four)**.** *Let $L = z_0 + u\Lambda$ be a unit triangular lattice with long direction u. There are offsets $s_b \in \mathbb{Z}/8$ with*

$$\chi\big(z_0 + u(a + b\omega)\big) = h(a + s_b), \qquad s_{b+1} - s_b \in \{4, 5\}.$$

Proof. By Theorem 2.2 and Rule A every run in the long direction has length exactly four, so each row is $a \mapsto h(a + s_b)$. The points a, $a+1$, $a+\omega$ form a unit triangle, so under a red run $t, \dots, t+3$ of row b the points $t, t+1, t+2$ of row $b+1$ are blue by (F); hence the blue run of row $b+1$ starts at $t-1$ or t, which says $s_{b+1} \in \{s_b + 4, s_b + 5\}$. □

Corollary 4.2 (four apart)**.** *In L, points z and $z+4u$ have opposite colours, since $h(x+4) \neq h(x)$.*

The differences $s_{b+1} - s_b$ are the *steps*; a *turn* is a step of 4 followed by 5 or of 5 followed by 4.

5. Rule B

Fix a row of L in its long direction. The points of the neighbouring rows sit above and below consecutive pairs of that row, and by (F) every such point under a monochromatic pair has the other colour. This fixes three of every four points in each neighbouring row; the fourth, above or below a change of colour, is *open*. Open points come in vertically aligned pairs, one above and one below the row (Figure 3).

Lemma 5.1. *The two points of an open pair have the same colour exactly when the two steps at this row form a turn.*

Proof. With the row as row 0 and offset 0, the point $a+\omega$ lies above the pair $a, a+1$, and $(a+1)-\omega$ lies below it at the same horizontal position. It is open exactly when $h(a) \neq h(a+1)$, and its colours are $h(a + d_1)$ and $h(a + 1 - d_0)$, where $d_0, d_1 \in \{4, 5\}$ are the steps below and above. For $a = 3$ these are $h(3 + d_1)$ and $h(4 - d_0)$, which agree iff $d_0 \neq d_1$; the case $a = 7$ is the same shifted by 4. □

Theorem 5.2 (Rule B)**.** *In an admissible colouring no unit triangular lattice has a turn. Equivalently, the two points of every open pair have different colours.*

Proof. Since $h(-x) = h(x + 3)$, the half-turn $z \mapsto -z$ replaces s_b by $3 - s_{-b}$ and reverses the order of the steps, so it suffices to exclude a step of 4 followed by 5. After an isometry and a translation along the rows, the lattice is Λ with long direction 1 and offsets $s_0 = 0$, $s_1 = 4$, $s_2 = 1$; rows $0, 1, 2$ are then fixed. We use only these three rows, rule (F), and Corollary 4.2 applied to $q\Lambda$.

Common deductions. Inside Λ alone, (F) gives five further colours (Table 1).

The cases. By Theorem 2.2 the long direction of $q\Lambda$ is $q\omega^j$ for some $j \in \{0, 1, 2\}$, and by Corollary 4.2 points of $q\Lambda$ four steps apart along it have opposite colours. In each case a few more deductions end in a monochromatic forbidden triangle:

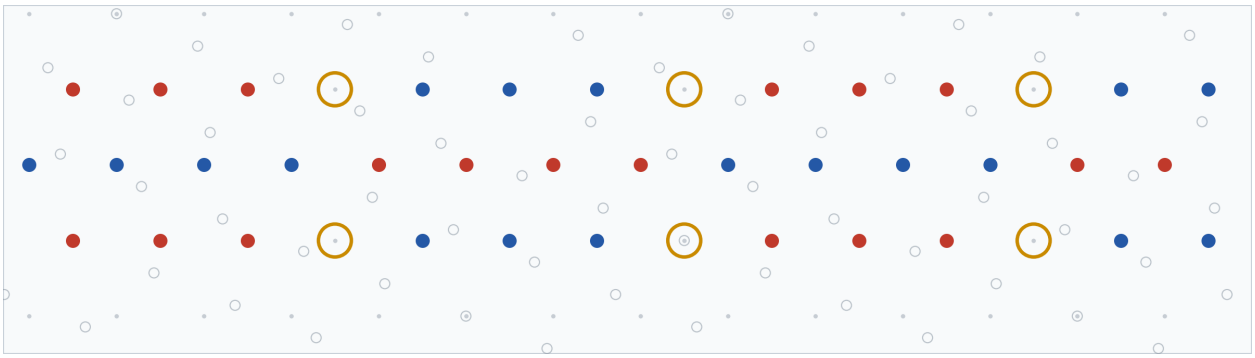


FIGURE 3. One row of the long direction, and the points it forces by unit triangles in the rows above and below. The open points (ringed) come in vertically aligned pairs.

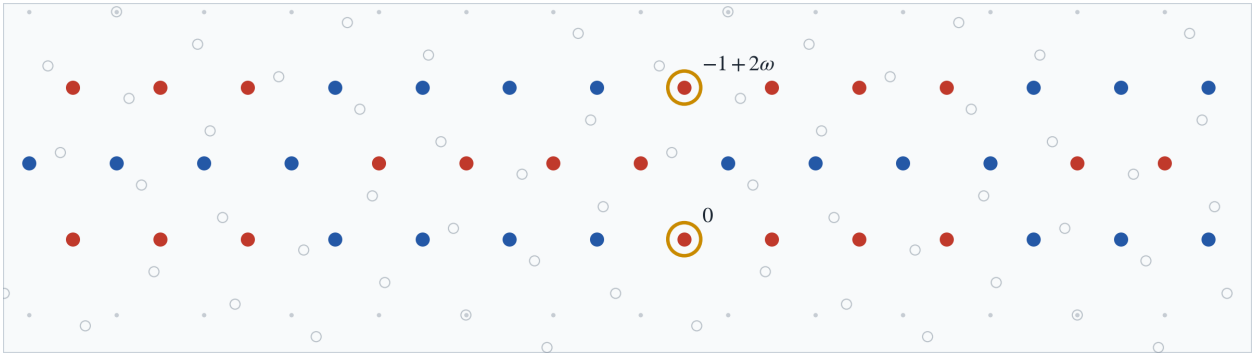


FIGURE 4. A same-coloured open pair: here 0 and $-1+2\omega$ are both red, and rows $0, 1, 2$ have offsets $0, 4, 1$, a turn.

TABLE 1. Deductions inside Λ, valid in every case.

	point	colour	because
1	$-2-\omega$	red	$T(1)$ with -3, -2 (blue)
2	$-1+3\omega$	blue	$T(1)$ with $-1+2\omega$, 2ω (red)
3	$2-\omega$	blue	$T(1)$ with 1, 2 (red)
4	$3-\omega$	blue	$T(1)$ with 2, 3 (red)
5	$3-2\omega$	red	$T(1)$ with $2-\omega$, $3-\omega$ (blue)

Case: the long direction of $q\Lambda$ is q.

	point	colour	because
1	$q(2-2\omega)$	red	four apart from $4+2\omega$ along q
2	$q(1+\omega)$	blue	four apart from $-2-\omega$ along q
3	$q(2+\omega)$	red	$T(1)$ with $-1+3\omega$, $q(1+\omega)$ (blue)

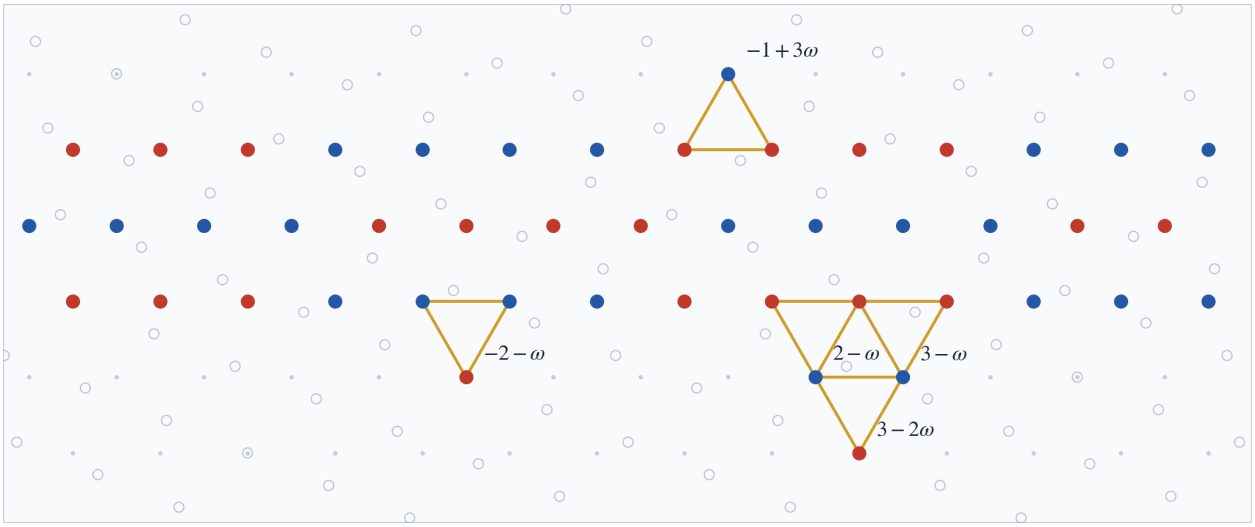


FIGURE 5. The five deductions of Table 1, inside Λ alone, from rows $0, 1, 2$.

Contradiction: $q(2+\omega)$, $q(2-2\omega)$, $3-2\omega$ are all red and form $T(1,3,4)$.

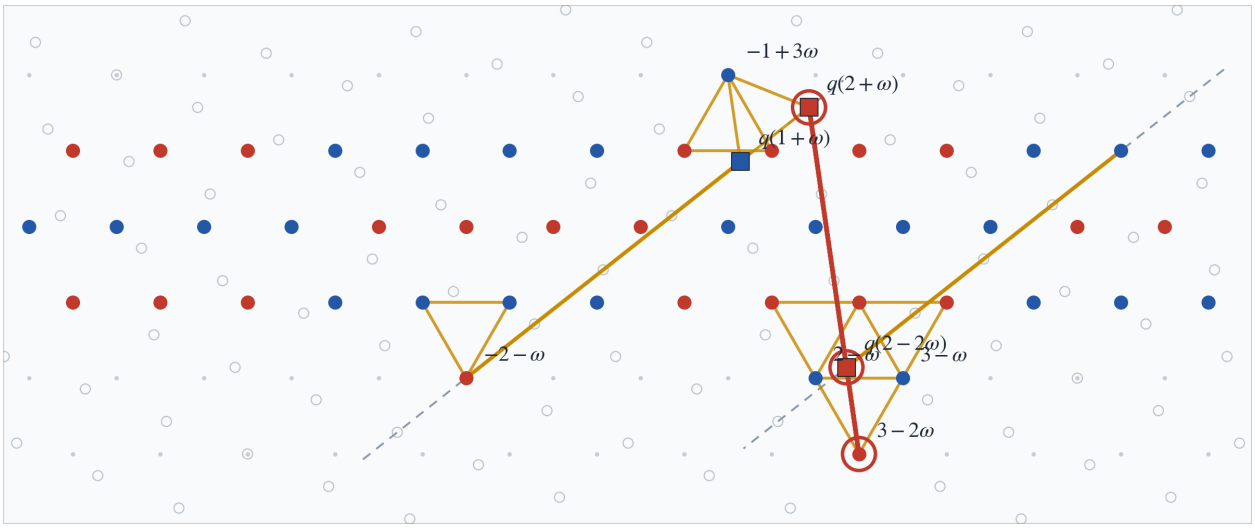


FIGURE 6. Rule B, long direction q: every deduction of this case (amber), the four-apart steps along the dashed lines, and the contradiction (red).

Case: the long direction of $q\Lambda$ is $q\omega$.

	point	colour	because
1	$q(1-2\omega)$	red	four apart from $-1+3\omega$ along $q\omega$
2	$q(2+\omega)$	blue	four apart from $3-2\omega$ along $q\omega$
3	$q(-2+\omega)$	blue	$T(1,3,4)$ with $q(1-2\omega)$, $3-2\omega$ (red)
4	$-q$	red	$T(1,3,4)$ with $-5+\omega$, $q(-2+\omega)$ (blue)
5	$q(-1+\omega)$	blue	$T(1)$ with $-q$, 0 (red)

Contradiction: $q(-2+\omega)$, $q(-1+\omega)$, $q(2+\omega)$ are all blue and form $T(1,3,4)$.

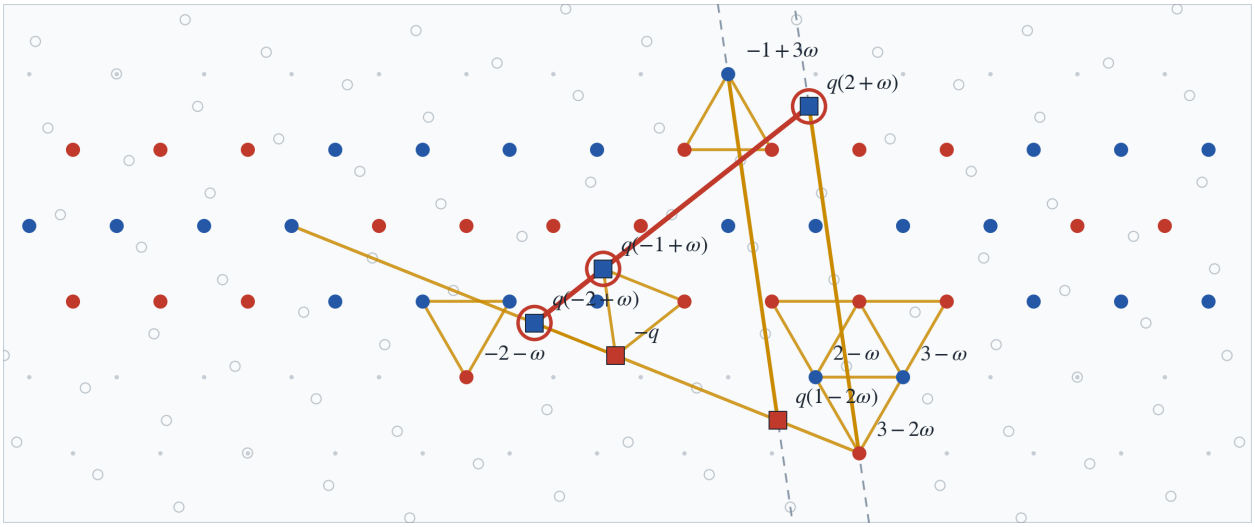


FIGURE 7. Rule B, long direction $q\omega$: every deduction of this case (amber), the four-apart steps along the dashed lines, and the contradiction (red).

Case: the long direction of $q\Lambda$ is $q\omega^2$.

	point	colour	because
1	$q(-4+\omega)$	blue	four apart from -7 along $q\omega^2$
2	$-q$	red	four apart from $-5+\omega$ along $q\omega^2$
3	$q(-4+4\omega)$	blue	four apart from 0 along $q\omega^2$
4	$q(-1+\omega)$	blue	$T(1)$ with $-q$, 0 (red)

Contradiction: $q(-4+4\omega)$, $q(-4+\omega)$, $q(-1+\omega)$ are all blue and form $T(3)$.

All three directions are impossible, so Λ has no turn. □

6. Distance eight and the proof of Theorem 1.1

Proposition 6.1. *In an admissible colouring, any two points at distance* 8 *have the same colour.*

Proof. Let $|p-p'|=8$ and take a unit triangular lattice in which $p'-p$ is eight unit steps. If its long direction is parallel to $p'-p$, the row through p,p' has period 8. Otherwise choose coordinates in which the long direction is ω times the old horizontal; since $8=\omega(8-8\omega)$, the endpoints have coordinates $(0,0)$ and $(8,-8)$ and colours $h(s_0)$ and $h(8+s_{-8})=h(s_{-8})$. By Rule B the eight steps between rows -8 and 0 are all equal to some $\delta\in\{4,5\}$, so $s_0-s_{-8}=8\delta\equiv 0 \pmod 8$. The other transverse direction is symmetric. □

Proof of Theorem 1.1. Suppose χ is admissible and $|x-y|=1$. The circles of radius 8 about x and y meet in a point r, and Proposition 6.1 gives $\chi(x)=\chi(r)=\chi(y)$. So every unit equilateral triangle is monochromatic, contradicting Lemma 2.1. □

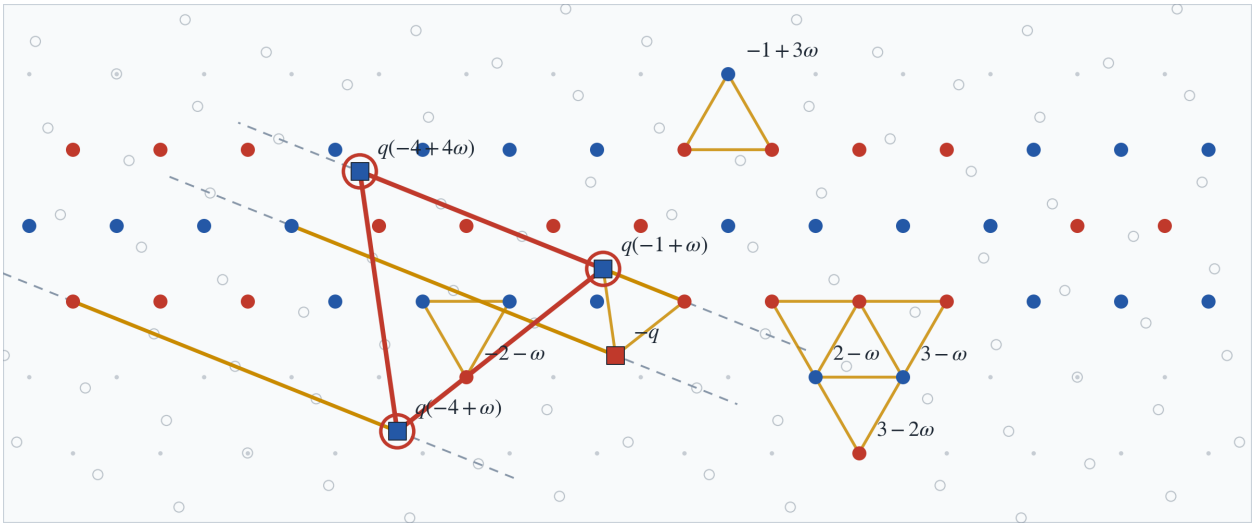


FIGURE 8. Rule B, long direction $q\omega^2$: every deduction of this case (amber), the four-apart steps along the dashed lines, and the contradiction (red).

7. Verification

The finite parts of Rule A and Rule B are recorded in certificates with standard-library verifiers. For Rule A the verifier checks that every point lies in $\Lambda \cup q\Lambda$, re-derives all forbidden triangles from coordinates, checks the symmetry statements and that the four cases' orbits cover all nine boundary colourings, and replays each case with rule (F). For Rule B it checks Lemma 5.1 for all four pairs of steps and every deduction of Table 1 and of the three cases, each against exactly one rule. The tables in this paper are generated from the certificates. Interactive walkthroughs of both rules accompany the code.

References

[1] P. Erdős, R. L. Graham, P. Montgomery, B. L. Rothschild, J. Spencer and E. G. Straus, *Euclidean Ramsey theorems. III*, in: Infinite and Finite Sets, Colloq. Math. Soc. János Bolyai 10, North-Holland, 1975, 559–583.

[2] G. Currier, K. Moore and C. H. Yip, *Any two-coloring of the plane contains monochromatic 3-term arithmetic progressions*, Combinatorica **44** (2024), no. 6, 1367–1380.

[3] A. Yan, *Monochromatic configurations in two-colored plane*, Bachelor's thesis, Charles University, Prague, 2025.

Appendix A. Forcing tables for Rule A

Points are named in Λ or as $q(m+n\omega)$. The row $-3,\dots,5$ is as in Lemma 3.2. In each case, deductions are listed before the case split and in each subcase; each subcase ends in a contradiction.

Case 1: blue $-\omega$, $3-\omega$, $-1+\omega$, $2+\omega$.

	point	colour	because
1	$-1-\omega$	red	$T(1)$ with -2, -1 (blue)
2	ω	blue	$T(1)$ with 0, 1 (red)
3	$1+\omega$	blue	$T(1)$ with 1, 2 (red)
4	$2-\omega$	blue	$T(1)$ with 1, 2 (red)
5	7	red	$T(1,3,4)$ with 3, 4 (blue)
6	2ω	red	$T(3)$ with $-\omega$, $3-\omega$ (blue)
7	$3-2\omega$	red	$T(1)$ with $2-\omega$, $3-\omega$ (blue)
8	$-1+2\omega$	red	$T(1)$ with $-1+\omega$, ω (blue)

	point	colour	because
9	$1+2\omega$	red	$T(1)$ with $1+\omega$, $2+\omega$ (blue)
10	$-1+3\omega$	blue	$T(1,3,4)$ with $-1-\omega$, $-1+2\omega$ (red)
11	$4+2\omega$	blue	$T(1,3,4)$ with 2ω, $1+2\omega$ (red)

Subcase: $q(2+\omega)$ blue.

	point	colour	because
1	$q(5-2\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(2+\omega)$ (blue)
2	$q(4-\omega)$	blue	$T(1,3,4)$ with $q(5-2\omega)$, 7 (red)
3	$q(3\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (blue)
4	$q(-\omega)$	red	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (blue)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all red and form $T(1,3,4)$.

Subcase: $q(2+\omega)$ red.

	point	colour	because
1	$q(2)$	blue	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (red)
2	$q(2-2\omega)$	blue	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (red)
3	$q(3)$	red	$T(1)$ with $q(2)$, $2+\omega$ (blue)
4	$q(5-2\omega)$	red	$T(1,3,4)$ with $q(2-2\omega)$, $4+2\omega$ (blue)
5	$q(3\omega)$	blue	$T(3)$ with 0, $q(3)$ (red)
6	$q(4-\omega)$	blue	$T(1,3,4)$ with $q(5-2\omega)$, 7 (red)

Contradiction: $q(3\omega)$, $-1+3\omega$, $q(4-\omega)$ are all blue and form $T(1,3,4)$.

Case 2: blue $3-\omega$, $-1+\omega$, $2+\omega$; red $-\omega$.

	point	colour	because
1	$-1-\omega$	red	$T(1)$ with -2, -1 (blue)
2	ω	blue	$T(1)$ with 0, 1 (red)
3	$1+\omega$	blue	$T(1)$ with 1, 2 (red)
4	$2-\omega$	blue	$T(1)$ with 1, 2 (red)
5	7	red	$T(1,3,4)$ with 3, 4 (blue)
6	$3-2\omega$	red	$T(1)$ with $2-\omega$, $3-\omega$ (blue)
7	$-1+2\omega$	red	$T(1)$ with $-1+\omega$, ω (blue)
8	$1+2\omega$	red	$T(1)$ with $1+\omega$, $2+\omega$ (blue)
9	$-1+3\omega$	blue	$T(1,3,4)$ with $-1-\omega$, $-1+2\omega$ (red)
10	2ω	red	$T(1)$ with ω, $1+\omega$ (blue)
11	$4+2\omega$	blue	$T(1,3,4)$ with 2ω, $1+2\omega$ (red)

Subcase: $q(2+\omega)$ blue.

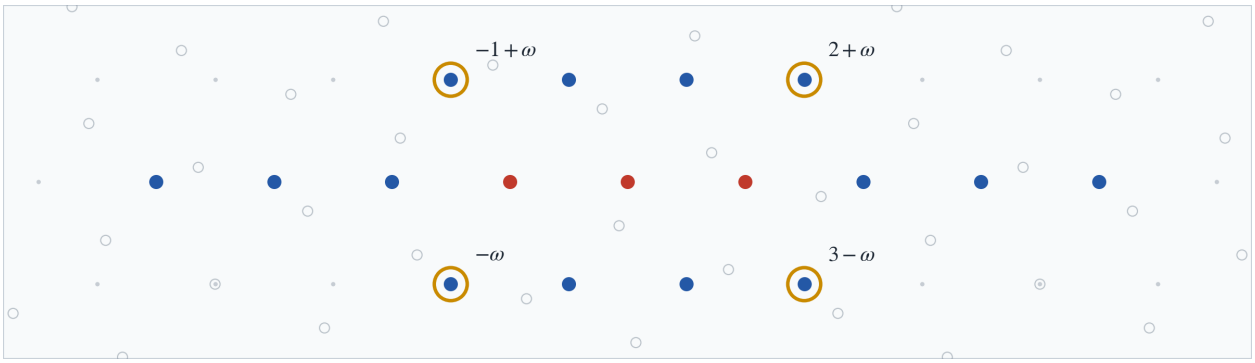


FIGURE 9. Rule A, case 1: the row, the four forced points, and the boundary colouring of this case (ringed).

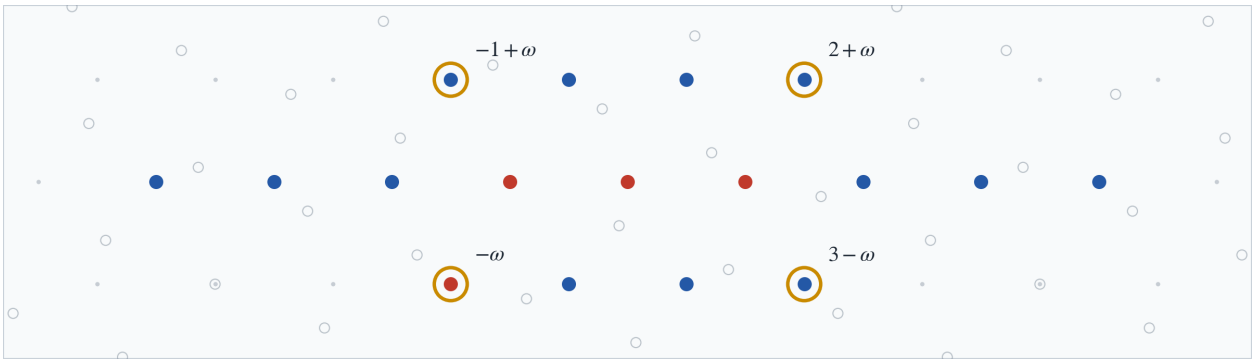


FIGURE 10. Rule A, case 2: the row, the four forced points, and the boundary colouring of this case (ringed).

	point	colour	because
1	$q(5-2\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(2+\omega)$ (blue)
2	$q(4-\omega)$	blue	$T(1,3,4)$ with $q(5-2\omega)$, 7 (red)
3	$q(3\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (blue)
4	$q(-\omega)$	red	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (blue)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all red and form $T(1,3,4)$.

Subcase: $q(2+\omega)$ red.

	point	colour	because
1	$q(2)$	blue	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (red)
2	$q(2-2\omega)$	blue	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (red)
3	$q(3)$	red	$T(1)$ with $q(2)$, $2+\omega$ (blue)
4	$q(5-2\omega)$	red	$T(1,3,4)$ with $q(2-2\omega)$, $4+2\omega$ (blue)
5	$q(3\omega)$	blue	$T(3)$ with 0, $q(3)$ (red)
6	$q(4-\omega)$	blue	$T(1,3,4)$ with $q(5-2\omega)$, 7 (red)

Contradiction: $q(3\omega)$, $-1+3\omega$, $q(4-\omega)$ are all blue and form $T(1,3,4)$.

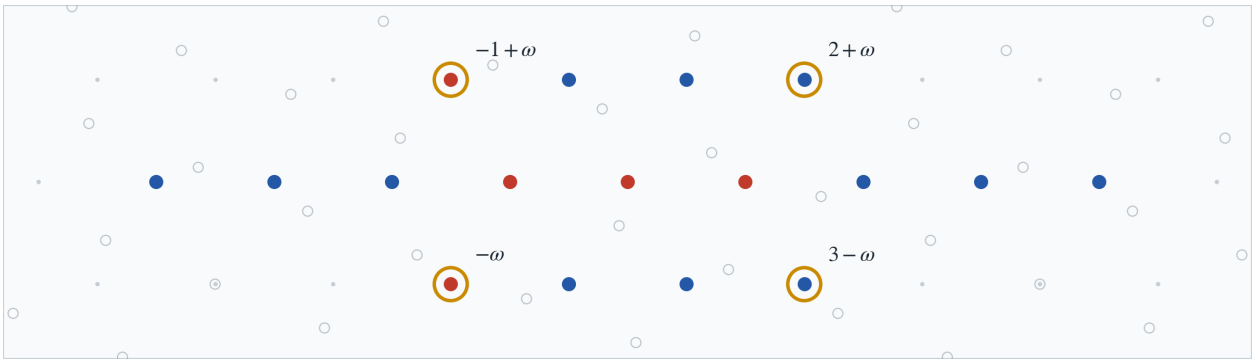


FIGURE 11. Rule A, case 3: the row, the four forced points, and the boundary colouring of this case (ringed).

Case 3: blue $3-\omega$, $2+\omega$; red $-\omega$, $-1+\omega$.

	point	colour	because
1	$3-4\omega$	red	$T(4)$ with -1, 3 (blue)
2	ω	blue	$T(1)$ with $-1+\omega$, 0 (red)
3	$1+\omega$	blue	$T(1)$ with 1, 2 (red)
4	7	red	$T(1,3,4)$ with 3, 4 (blue)
5	$5-\omega$	red	$T(1)$ with 4, 5 (blue)
6	$4-5\omega$	blue	$T(1,3,4)$ with $-\omega$, $3-4\omega$ (red)
7	$1+2\omega$	red	$T(1)$ with $1+\omega$, $2+\omega$ (blue)
8	2ω	red	$T(1)$ with ω, $1+\omega$ (blue)
9	$4+2\omega$	blue	$T(1,3,4)$ with 2ω, $1+2\omega$ (red)

Subcase: $q(5-5\omega)$ blue.

	point	colour	because
1	$q(1-\omega)$	red	$T(4)$ with $4-5\omega$, $q(5-5\omega)$ (blue)
2	$q(4-5\omega)$	red	$T(1,3,4)$ with $4-5\omega$, $q(5-5\omega)$ (blue)
3	$q(-\omega)$	blue	$T(1)$ with 0, $q(1-\omega)$ (red)
4	$q(4-\omega)$	blue	$T(4)$ with $q(4-5\omega)$, 7 (red)

Contradiction: $q(-\omega)$, $2+\omega$, $q(4-\omega)$ are all blue and form $T(1,3,4)$.

Subcase: $q(5-5\omega)$ red.

	point	colour	because
1	$q(5-\omega)$	blue	$T(1,3,4)$ with $5-\omega$, $q(5-5\omega)$ (red)
2	$q(5-2\omega)$	blue	$T(3)$ with $q(5-5\omega)$, 7 (red)

Contradiction: $q(5-\omega)$, $q(5-2\omega)$, $4+2\omega$ are all blue and form $T(1)$.

Case 4: blue $-\omega$, $2+\omega$; red $3-\omega$, $-1+\omega$.

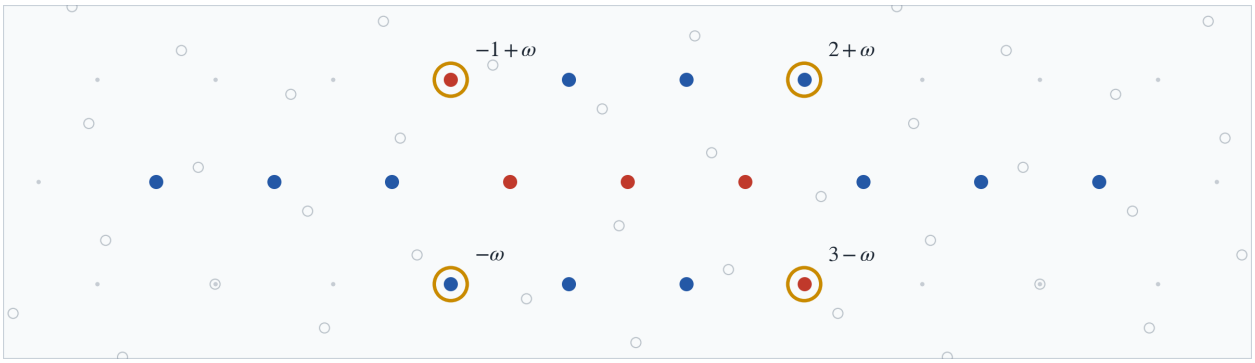


FIGURE 12. Rule A, case 4: the row, the four forced points, and the boundary colouring of this case (ringed).

	point	colour	because
1	$3-4\omega$	red	$T(1,3,4)$ with -1, $-\omega$ (blue)
2	ω	blue	$T(1)$ with $-1+\omega$, 0 (red)
3	$3-3\omega$	blue	$T(1,3,4)$ with $-1+\omega$, 0 (red)
4	$1+\omega$	blue	$T(1)$ with 1, 2 (red)
5	$-1+3\omega$	blue	$T(1,3,4)$ with 2, $3-\omega$ (red)
6	$6-4\omega$	blue	$T(1,3,4)$ with 2, $3-\omega$ (red)
7	7	red	$T(1,3,4)$ with 3, 4 (blue)
8	$5-\omega$	red	$T(1)$ with 4, 5 (blue)
9	$1+2\omega$	red	$T(1)$ with $1+\omega$, $2+\omega$ (blue)
10	2ω	red	$T(1)$ with ω, $1+\omega$ (blue)
11	$4+2\omega$	blue	$T(1,3,4)$ with 2ω, $1+2\omega$ (red)

Subcase: $q(4-\omega)$ blue.

	point	colour	because
1	$q(3\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (blue)
2	$q(-\omega)$	red	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (blue)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all red and form $T(1,3,4)$.

Subcase: $q(4-\omega)$ red.

	point	colour	because
1	$q(5-2\omega)$	blue	$T(1,3,4)$ with $q(4-\omega)$, 7 (red)
2	$q(4-5\omega)$	blue	$T(4)$ with $q(4-\omega)$, 7 (red)
3	$q(2+\omega)$	red	$T(1,3,4)$ with $-1+3\omega$, $q(5-2\omega)$ (blue)
4	$q(1-2\omega)$	red	$T(4)$ with $-1+3\omega$, $q(5-2\omega)$ (blue)
5	$q(2-2\omega)$	red	$T(1,3,4)$ with $q(5-2\omega)$, $4+2\omega$ (blue)
6	$q(5-\omega)$	red	$T(1)$ with $q(5-2\omega)$, $4+2\omega$ (blue)
7	$q(6-3\omega)$	red	$T(1)$ with $q(5-2\omega)$, $4+2\omega$ (blue)
8	$q(3-5\omega)$	red	$T(1)$ with $q(4-5\omega)$, $6-4\omega$ (blue)
9	$q(3)$	blue	$T(1,3,4)$ with $q(2+\omega)$, $q(6-3\omega)$ (red)
10	$q(1-\omega)$	blue	$T(1)$ with $q(1-2\omega)$, $q(2-2\omega)$ (red)

	point	colour	because
11	$q(5-5\omega)$	blue	$T(1,3,4)$ with $q(5-\omega)$, $5-\omega$ (red)
12	$q(3-4\omega)$	red	$T(1,3,4)$ with $q(3)$, $2+\omega$ (blue)
13	$4-5\omega$	red	$T(4)$ with $q(1-\omega)$, $q(5-5\omega)$ (blue)
14	$7-7\omega$	blue	$T(1,3,4)$ with $q(3-4\omega)$, $q(3-5\omega)$ (red)
15	$4-4\omega$	blue	$T(1)$ with $3-4\omega$, $4-5\omega$ (red)

Contradiction: $3-3\omega$, $4-4\omega$, $7-7\omega$ are all blue and form $T(1,3,4)$.

Appendix B. Rule A in two splits

Here $3-\omega$ is blue, by σ and Lemma 3.3, and the first split is at q.

	point	colour	because
1	-6	red	$T(1,3,4)$ with -3, -2 (blue)
2	$-3+\omega$	red	$T(1)$ with -3, -2 (blue)
3	$-2-\omega$	red	$T(1)$ with -3, -2 (blue)
4	-5	red	$T(1,3,4)$ with -2, -1 (blue)
5	$-2+\omega$	red	$T(1)$ with -2, -1 (blue)
6	$-1-\omega$	red	$T(1)$ with -2, -1 (blue)
7	$-1+4\omega$	red	$T(4)$ with -1, 3 (blue)
8	ω	blue	$T(1)$ with 0, 1 (red)
9	$1+\omega$	blue	$T(1)$ with 1, 2 (red)
10	$2-\omega$	blue	$T(1)$ with 1, 2 (red)
11	$3-2\omega$	red	$T(1)$ with $2-\omega$, $3-\omega$ (blue)
12	$3+\omega$	red	$T(1)$ with 3, 4 (blue)
13	7	red	$T(1,3,4)$ with 3, 4 (blue)
14	$4+\omega$	red	$T(1)$ with 4, 5 (blue)
15	$5-\omega$	red	$T(1)$ with 4, 5 (blue)
16	$-5-\omega$	blue	$T(1)$ with -6, -5 (red)
17	$-3+2\omega$	blue	$T(1)$ with $-3+\omega$, $-2+\omega$ (red)
18	$-1-2\omega$	blue	$T(1)$ with $-2-\omega$, $-1-\omega$ (red)
19	2ω	red	$T(1)$ with ω, $1+\omega$ (blue)
20	$3+2\omega$	blue	$T(1,3,4)$ with $3-2\omega$, $3+\omega$ (red)

Case: q blue.

	point	colour	because
1	$q(-3+4\omega)$	red	$T(1,3,4)$ with $-3+2\omega$, q (blue)
2	$q(1+3\omega)$	red	$T(3)$ with $-3+2\omega$, q (blue)
3	$q(2-\omega)$	red	$T(1,3,4)$ with $-3+2\omega$, q (blue)
4	$q(\omega)$	blue	$T(3)$ with $q(-3+4\omega)$, $-2-\omega$ (red)
5	$q(5-\omega)$	blue	$T(3)$ with $q(2-\omega)$, $5-\omega$ (red)
6	$q(4-3\omega)$	red	$T(1,3,4)$ with $q(\omega)$, q (blue)
7	$q(5-3\omega)$	blue	$T(1)$ with $q(4-3\omega)$, $5-\omega$ (red)

Subcase: $q(4-\omega)$ blue.

	point	colour	because
1	$q(1-\omega)$	red	$T(1,3,4)$ with $q(4-\omega)$, $q(5-\omega)$ (blue)
2	$q(-3+3\omega)$	blue	$T(4)$ with $q(1+3\omega)$, $q(1-\omega)$ (red)
3	$q(-2-\omega)$	blue	$T(1,3,4)$ with $q(1-\omega)$, $q(2-\omega)$ (red)
4	$q(-\omega)$	blue	$T(1)$ with 0, $q(1-\omega)$ (red)
5	$q(2-2\omega)$	blue	$T(1)$ with $q(1-\omega)$, $q(2-\omega)$ (red)
6	$q(-6+3\omega)$	red	$T(1,3,4)$ with $q(-3+3\omega)$, $-3+2\omega$ (blue)
7	$2+\omega$	red	$T(1,3,4)$ with $q(-\omega)$, $q(4-\omega)$ (blue)
8	$-1+\omega$	blue	$T(1,3,4)$ with $-2+\omega$, $2+\omega$ (red)
9	$q(-1-\omega)$	blue	$T(1,3,4)$ with $q(2-\omega)$, $2+\omega$ (red)
10	$q(3-2\omega)$	blue	$T(1)$ with $q(2-\omega)$, $2+\omega$ (red)
11	$-4+\omega$	red	$T(3)$ with $-1-2\omega$, $-1+\omega$ (blue)
12	$1-3\omega$	red	$T(1)$ with $q(-2-\omega)$, $q(-1-\omega)$ (blue)
13	$4+2\omega$	red	$T(1,3,4)$ with $q(2-2\omega)$, $q(3-2\omega)$ (blue)
14	$-7+\omega$	blue	$T(1,3,4)$ with $-4+\omega$, $-3+\omega$ (red)
15	$-3-3\omega$	blue	$T(4)$ with $-3+\omega$, $1-3\omega$ (red)
16	$1+2\omega$	blue	$T(1,3,4)$ with 2ω, $4+2\omega$ (red)
17	$-4-2\omega$	red	$T(1,3,4)$ with $-7+\omega$, $-3-3\omega$ (blue)
18	$-2+2\omega$	red	$T(1,3,4)$ with $-3+2\omega$, $1+2\omega$ (blue)
19	$q(-6+6\omega)$	blue	$T(1,3,4)$ with $q(-6+3\omega)$, $-4-2\omega$ (red)
20	$-5+2\omega$	blue	$T(3)$ with $-2-\omega$, $-2+2\omega$ (red)
21	$-8+3\omega$	red	$T(1,3,4)$ with $q(-6+6\omega)$, $q(-3+3\omega)$ (blue)
22	$-5+3\omega$	red	$T(1,3,4)$ with $-5-\omega$, $-5+2\omega$ (blue)

Contradiction: $-8+3\omega$, -5, $-5+3\omega$ are all red and form $T(3)$.

Subcase: $q(4-\omega)$ red.

	point	colour	because
1	$q(5-2\omega)$	blue	$T(1,3,4)$ with $q(4-\omega)$, 7 (red)
2	$4+2\omega$	red	$T(1)$ with $q(5-\omega)$, $q(5-2\omega)$ (blue)
3	$q(4-2\omega)$	red	$T(1)$ with $q(5-2\omega)$, $q(5-3\omega)$ (blue)
4	$1+2\omega$	blue	$T(1,3,4)$ with 2ω, $4+2\omega$ (red)
5	$2+\omega$	blue	$T(1)$ with $q(4-\omega)$, $q(4-2\omega)$ (red)

Contradiction: $1+\omega$, $1+2\omega$, $2+\omega$ are all blue and form $T(1)$.

Case: q red.

	point	colour	because
1	$q(\omega)$	blue	$T(1)$ with 0, q (red)
2	$q(1-\omega)$	blue	$T(1)$ with 0, q (red)
3	$q(2-\omega)$	blue	$T(1,3,4)$ with q, $5-\omega$ (red)
4	$q(4-3\omega)$	blue	$T(1,3,4)$ with q, $5-\omega$ (red)
5	$1+4\omega$	red	$T(4)$ with $q(\omega)$, $q(4-3\omega)$ (blue)
6	$q(2-2\omega)$	red	$T(1)$ with $q(1-\omega)$, $q(2-\omega)$ (blue)
7	$q(-1+2\omega)$	red	$T(1,3,4)$ with $-3+2\omega$, $q(2-\omega)$ (blue)
8	$q(2+3\omega)$	red	$T(4)$ with $-3+2\omega$, $q(2-\omega)$ (blue)

	point	colour	because
9	$-3+4\omega$	blue	$T(4)$ with 1, $1+4\omega$ (red)
10	5ω	blue	$T(1,3,4)$ with $1+4\omega$, $4+\omega$ (red)
11	$q(1-2\omega)$	blue	$T(1)$ with $q(2-2\omega)$, $3-2\omega$ (red)
12	$-4+5\omega$	red	$T(1,3,4)$ with $-3+4\omega$, ω (blue)
13	$-1+6\omega$	red	$T(1,3,4)$ with 5ω, $3+2\omega$ (blue)
14	$-1+3\omega$	red	$T(1,3,4)$ with $q(1-\omega)$, $q(1-2\omega)$ (blue)
15	$q(-\omega)$	red	$T(1)$ with $q(1-\omega)$, $q(1-2\omega)$ (blue)
16	$q(-4+5\omega)$	blue	$T(3)$ with $-4+5\omega$, $q(-1+2\omega)$ (red)
17	$-1+7\omega$	blue	$T(1,3,4)$ with $-1+3\omega$, $-1+6\omega$ (red)
18	$q(3\omega)$	blue	$T(1,3,4)$ with 0, $q(-\omega)$ (red)
19	$q(-3+4\omega)$	red	$T(1,3,4)$ with $q(-4+5\omega)$, $q(\omega)$ (blue)
20	$q(5\omega)$	red	$T(4)$ with $q(-4+5\omega)$, $q(\omega)$ (blue)
21	$q(1+4\omega)$	blue	$T(4)$ with $q(-3+4\omega)$, q (red)
22	$q(4\omega)$	blue	$T(1)$ with $-4+5\omega$, $q(5\omega)$ (red)
23	$q(1+3\omega)$	red	$T(1)$ with $q(4\omega)$, $q(1+4\omega)$ (blue)
24	$-5+8\omega$	red	$T(1,3,4)$ with $q(4\omega)$, $q(3\omega)$ (blue)
25	7ω	blue	$T(1,3,4)$ with $q(1+3\omega)$, $q(2+3\omega)$ (red)
26	$-4+7\omega$	blue	$T(1,3,4)$ with $-5+8\omega$, $-1+4\omega$ (red)

Contradiction: $-4+7\omega$, $-1+7\omega$, 7ω are all blue and form $T(1,3,4)$.